FARUK HASAN

Senior AI Test Automation Engineer | LLM Evaluation | Agentic AI & Quality Engineering

Long Island, New York | Email: fhasan.mohammad@gmail.com | **Mobile:** (631) 480-7466
LinkedIn | **Portfolio:** https://faruk-hasan.com/ | Github

PROFESSIONAL SUMMARY

Senior QA/SDET with 6+ years of test automation experience across web, API, and cloud applications. Builds scalable Playwright/TypeScript and Python automation and integrates quality checks into CI/CD workflows. Current AI quality work includes DeepEval-based evaluation, deterministic and LLM-as-a-judge checks, prompt/response validation, and evaluation of agentic testing workflows using MCP servers, Playwright MCP, Playwright Agents, GitHub Copilot, and Claude. Strong background in production defect analysis, regression strategy, and translating complex workflows into measurable quality criteria.

TECHNICAL EXPERTISE

AI & LLM Quality Engineering: DeepEval, LLM-as-a-judge, deterministic evaluation, prompt/response validation, relevance and groundedness evaluation, AI-assisted test automation

Agentic AI & MCP: Model Context Protocol (MCP), application MCP servers, Playwright MCP, Playwright Agents, tool-call/workflow validation, GitHub Copilot, Claude, OpenAI API

Automation: Playwright (TypeScript/Python), Pytest, Selenium WebDriver (Java/Python), Appium, TestNG, JUnit, Cucumber, Page Object Model, fixtures, data-driven testing

API / Data / Performance: Postman, REST API testing, SQL, MySQL, MongoDB, JMeter

CI/CD & Cloud: TeamCity, Jenkins, GitHub Actions, Docker, AWS (EC2, S3), Git/GitHub, shell scripting

Programming & Web: Python, TypeScript/JavaScript, Java, SQL, HTML/CSS, React, Vite

Quality & Observability: Jira, Zephyr, TestRail, Allure, Grafana, production defect analysis, system logs, release-readiness metrics

PROFESSIONAL EXPERIENCE

Digital.ai, Remote **September 2022 – Present**

Role: Senior Software QA Engineer

- **AI & LLM Evaluation:** Apply DeepEval and Python-based evaluation techniques to AI-assisted QA workflows, combining deterministic checks with LLM-as-a-judge evaluation for response relevance, correctness, groundedness, completeness, and overall quality.
- **Agentic AI / MCP Quality Engineering:** Evaluate engineering workflows involving application MCP servers, Playwright MCP, Playwright Agents, GitHub Copilot, and Claude; assess tool selection, tool sequencing, generated test behavior, edge cases, and end-to-end task outcomes.
- **Playwright Automation Architecture:** Develop and maintain a scalable Playwright/TypeScript quality platform supporting 300+ automated tests with reusable fixtures, Page Object Models, API-driven setup, authentication/session management, and data-driven testing.

- **AI-Assisted Test Development:** Use LLMs for test generation, prompt testing, response validation, exploratory analysis, and automation workflows while separating exact deterministic assertions from semantic AI evaluation.
- **Delivery Efficiency:** Reduced UI regression execution time by 50% through parallel execution and cross-browser testing, enabling faster CI feedback, earlier defect detection, and more reliable releases.
- **CI/CD & Release Enablement:** Build and maintain continuous-testing workflows in TeamCity and Jenkins using shell scripting and AWS to support automated regression, quality gates, and release readiness.
- **Production Quality & Regression:** Investigate production issues, reproduce failures, validate fixes, and convert findings into targeted Playwright, Python, API, and performance regression coverage.
- **Technical Leadership:** Share Playwright, automation, MCP, and emerging AI-testing practices with the QE Guild; mentor interns and junior engineers on framework design, quality engineering, and test strategy.

M.M. Hayes, Co., Inc. NY **March 2020 – August 2022**
Role: Software Test Engineer

- **Automation Framework:** Developed and maintained a Selenium WebDriver framework using Java and TestNG with reusable components and automated regression coverage for critical business workflows.
- **Backend & Integration Validation:** Implemented SQL-based validation to verify application data, business rules, and end-to-end transaction integrity across integrated systems.
- **Release Readiness:** Improved regression efficiency through parallel execution, cross-browser testing, and optimized test selection for faster release feedback.
- **Mobile / POS / Desktop Quality:** Built Appium/Java mobile automation and executed automated and manual testing for POS and desktop applications across business-critical workflows.
- **Traceability & Mentorship:** Established requirements traceability in Jira and mentored junior engineers on Selenium, Java, automation design, and QA best practices.

AI & AUTOMATION PROJECTS

AI Learning Products & Machine Learning Course | Udemy
Designed and launched a 9+ hour hands-on AI/ML learning product covering Python, NLP, Scikit-learn, Naive Bayes, data preparation, and development of a machine-learning chatbot. **Udemy Course**

Playwright Test Results Extension | Chrome Web Store
Built a JavaScript Chrome extension that turns raw Playwright test output into an accessible real-time results experience, helping developers quickly inspect test status and failures. **Store Link: Chrome Web Store**

EDUCATION

Master of Science in Electrical Engineering *(Graduated: 2016)*
Loyola Marymount University

Bachelor of Science in Electrical Engineering *(Graduated: 2011)*
Queen Mary, University of London